1 What a displayed formula shows

(a) State what a displayed formula must show. [2]

(b) Explain how a double bond is represented in a displayed formula. [1]

[3]

---

2 Counting atoms

A: CH_3CH_2OH B: CH_3COOH

(a) State the number of hydrogen atoms in molecule A. [1]

(b) Give the molecular formula of molecule B. [1]

[2]

---

3 Interpreting a formula

(a) Write the molecular formula of $CH_3CH_2CH_2OH$. [1]

(b) State the total number of atoms in one molecule of this compound. [1]

[2]

---

4 Recognising complete formulae

(a) Explain why CH_3CH_3 is not a displayed formula. [1]

(b) Draw the displayed formula of CH_3CH_3. [2]

[3]

5 Drawing an alkene

(a) Draw the displayed formula of $CH_2=CH_2$. [2]

(b) Count the C-H bonds and the C=C bonds in your drawing. [2]

[4]

---

6 Drawing an alcohol

(a) Draw the displayed formula of CH_3CH_2OH. [3]

(b) Circle the functional group on your drawing. [1]

[4]

---

7 Drawing a carboxylic acid

(a) Draw the displayed formula of CH_3COOH. [3]

(b) State how many single bonds and double bonds your drawing contains. [2]

[5]

8 Functional groups

(a) Explain what is meant by a functional group. [2]

(b) State why compounds with the same functional group have similar reactions. [1]

[3]

---

9 Identifying groups

(a) Give the functional group present in alcohols. [1]

(b) Give the functional group present in carboxylic acids. [1]

[2]

---

10 Classifying compounds

A: CH_3CH_3 B: $CH_2=CH_2$ C: CH_3OH D: CH_3COOH

(a) State the homologous series of A and B. [2]

(b) State the homologous series of C and D. [2]

[4]

---

11 Finding a functional group

P: CH_3CH_2OH Q: CH_3CH_2COOH

(a) Identify the alcohol: P or Q. [1]

(b) Write the functional group of the alcohol. [1]

[2]

12 General formulae of hydrocarbons

(a) State the general formula of an alkane. [1]

(b) State the general formula of an alkene. [1]

[2]

---

13 Using general formulae

(a) Deduce the molecular formula of an alkane with five carbon atoms. [1]

(b) Deduce the molecular formula of an alkene with five carbon atoms. [1]

[2]

---

14 Completing a sequence Extended

CH_4, C_2H_6, ..., C_4H_{10}, ...

(a) Complete the two missing formulae in the alkane sequence. [2]

(b) State the change in composition from each member to the next. [1]

[3]

---

15 Choosing a formula

Choose from: C_6H_{12}, C_6H_{14}, $C_6H_{13}OH$

(a) Select the formula that can represent an alkane. [1]

(b) Justify your selection using a general formula. [2]

[3]

16 The alcohol series

(a) State the general formula of an alcohol using n for its carbon number. [1]

(b) Use your formula to deduce the molecular formula when n = 4. [1]

[2]

---

17 The carboxylic acid series

Use the general formula $C_nH_{2n+1}COOH$.

(a) Find the molecular formula when n = 2. [2]

(b) State the total number of carbon atoms in this molecule. [1]

[3]

---

18 Working backwards

An alkane molecule contains 18 hydrogen atoms.

(a) Calculate the number of carbon atoms in the alkane. [2]

(b) Write its molecular formula. [1]

[3]

---

19 Comparing formulae

(a) Give the molecular formula of an alkene containing eight carbon atoms. [1]

(b) State how its hydrogen count compares with the eight-carbon alkane. [1]

[2]

20 Saturated compounds

(a) Define saturated in terms of carbon-carbon bonds. [1]

(b) Classify $CH_3CH_2CH_3$ as saturated or unsaturated. Give a reason. [2]

[3]

---

21 Unsaturated compounds

(a) Define unsaturated in terms of carbon-carbon bonds. [1]

(b) Classify $CH_3CH{=}CH_2$ as saturated or unsaturated. Give a reason. [2]

[3]

---

22 Using the saturation definition

(a) State whether CH_3CH_2OH is saturated. [1]

(b) State the type of carbon-carbon bond in CH_3CH_2OH. [1]

[2]

---

23 Spotting a double bond

A: $CH_3CH_2CH_3$ B: $CH_3CH{=}CH_2$

(a) Choose the unsaturated compound: A or B. [1]

(b) Copy the carbon-carbon double bond symbol from your choice. [1]

[2]

24 A family of compounds

(a) Complete: a homologous series is a family of ..... compounds. [1]

(b) State why members have similar chemical properties. [1]

[2]

---

25 Same functional group

A: CH_3OH B: CH_3CH_2OH

(a) State the functional group shared by A and B. [1]

(b) State whether A and B have similar chemical properties. [1]

[2]

---

26 Simple trends Extended

The table shows three successive alkanes.

Carbon atoms	1	2	3
Boiling point / °C	-162	-89	-42

(a) State whether boiling point rises or falls as carbon number increases. [1]

(b) Predict whether the next alkane has a higher or lower boiling point. [1]

[2]

27 General formula matching

Choose from: C_nH_{2n+2}, C_nH_{2n}, $C_nH_{2n+1}OH$

(a) Copy the general formula for alkanes. [1]

(b) Copy the general formula for alcohols. [1]

[2]

---

28 An acid formula

(a) Count the carbon atoms in CH_3CH_2COOH. [1]

(b) Draw its displayed formula. [3]

[4]

---

29 An alcohol formula

(a) Count the hydrogen atoms in CH_3OH. [1]

(b) Draw its displayed formula. [2]

[3]

30 Structural formulae Extended

(a) State what a structural formula tells you about atom arrangement. [1]

(b) Count the hydrogen atoms in CH_3CH_2OH. [1]

[2]

---

31 The meaning of isomers Extended

(a) Complete: structural isomers have the same ..... formula. [1]

(b) Complete: they have different ..... formulae. [1]

[2]

---

32 Recognising an isomer pair Extended

A: $CH_3CH_2CH_2CH_3$ B: $CH_3CH(CH_3)CH_3$

(a) Count the carbon and hydrogen atoms in A. [2]

(b) State whether A and B have the same molecular formula. [1]

[3]

33 Quick formula check

(a) Give the molecular formula of an alkane with three carbon atoms. [1]

(b) Give the molecular formula of an alkene with three carbon atoms. [1]

[2]

---

34 The role of a functional group

(a) Give the functional group of a carboxylic acid. [1]

(b) State what a functional group determines about a compound. [1]

[2]

---

35 Saturation review

(a) State the meaning of saturated. [1]

(b) State the meaning of unsaturated. [1]

[2]

---

36 Reading an alcohol formula

(a) Give the molecular formula of CH_3CH_2OH. [1]

(b) State how many O-H bonds are present in one molecule. [1]

[2]